Electrophysical Agents:

ES, TENS, CRYOTHERAPY, ULTRASOUND

Irvind Sihota

Learning outcomes

- Briefly revise the aims of acute injury management
- Explain the use of the following for acute injury rehabilitation:
  - Electrical stimulation (ES)
  - Transcutaneous electrical neuromuscular stimulation (TENS)
  - Cryotherapy
  - Ultrasound (US)
- of acute pain in musculoskeletal practice. Identify the approaches used in physiotherapy for the management

Acute management of pain

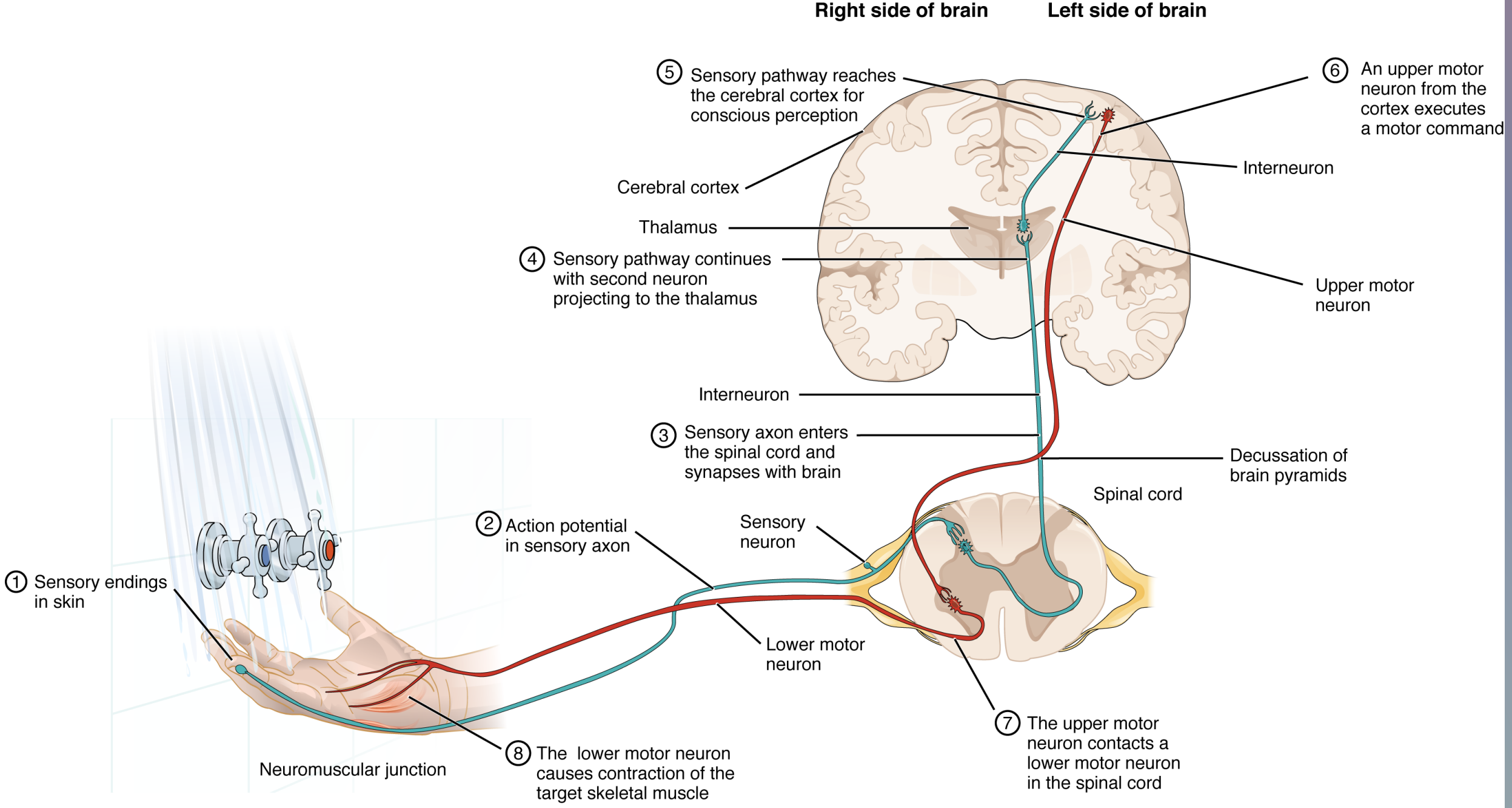
General physiotherapy aims of early phase management (Phase I & II)

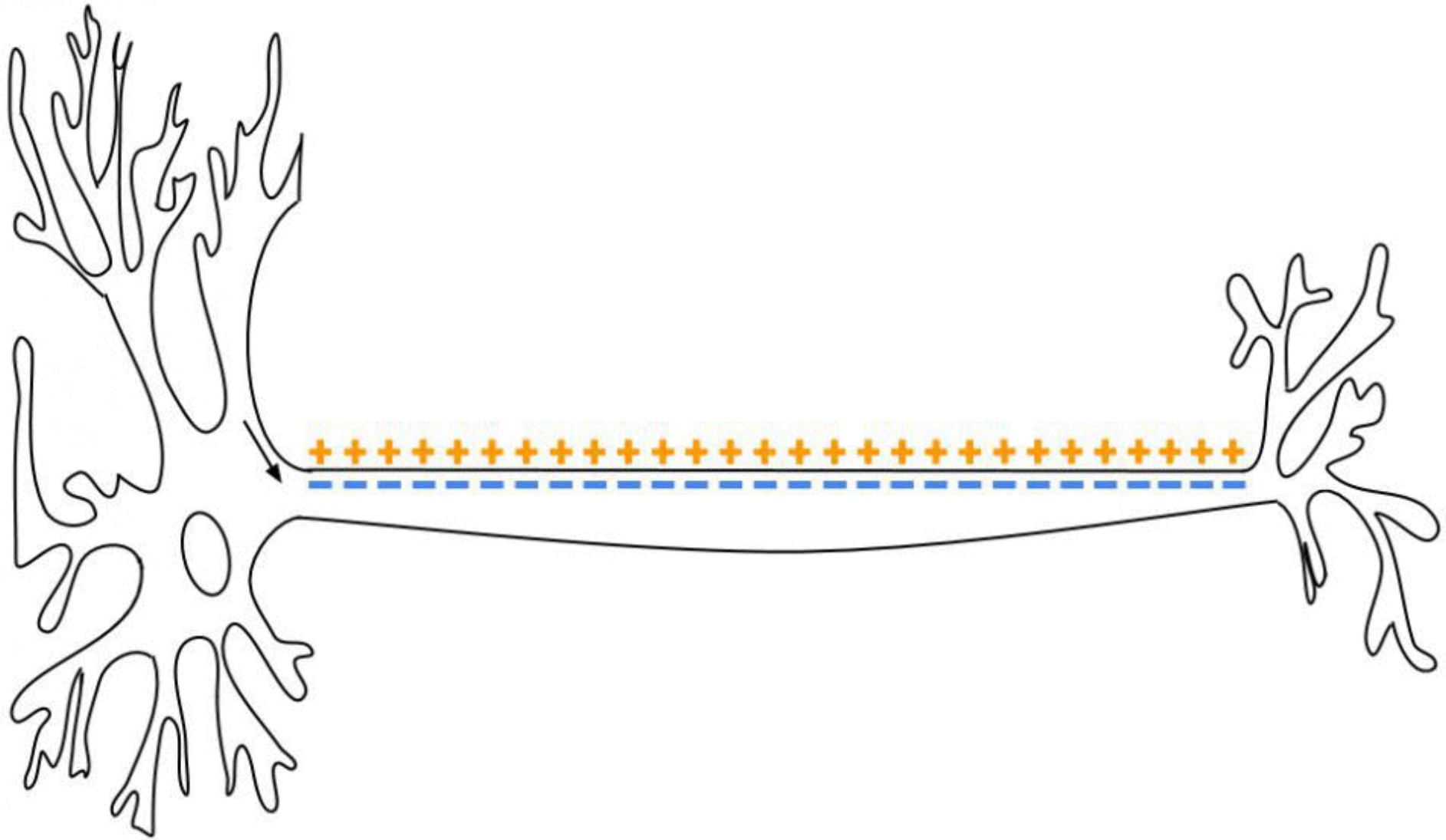
PHASE 1: BLEEDING & PHASE 2: INFLAMMATION

The aims are:

- to reduce pain;
- to limit and reduce inflammatory exudates;
- to reduce metabolic demands of tissue;
- to protect newly damaged tissue from further injury;
- to protect the newly-forming tissue from disruption;
- to promote new tissue growth and fibre realignment;
- to maintain general levels of cardiovascular and musculoskeletal fitness/activity.

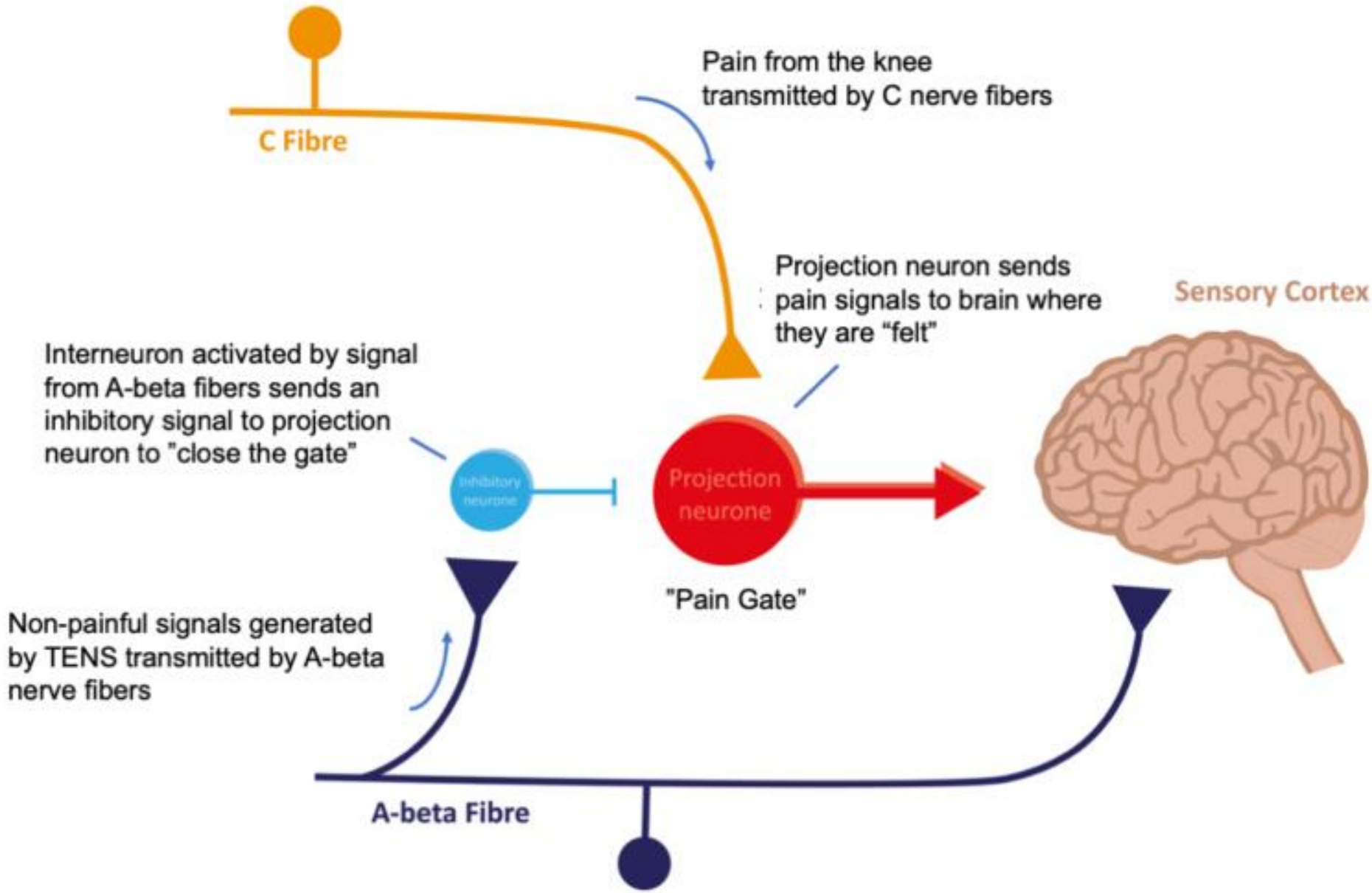
Porter, Stuart. Tidy's Physiotherapy E-Book, Elsevier Health Sciences, 2013. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/londonmet/detail.action?docID=1722589


Right side of brain
Left side of brain
5 Sensory pathway reaches the cerebral cortex for conscious perception
6 An upper motor neuron from the cortex executes a motor command
Cerebral cortex
Interneuron
Thalamus
4 Sensory pathway continues with second neuron projecting to the thalamus
Upper motor neuron
Interneuron
3 Sensory axon enters the spinal cord and synapses with brain
Decussation of brain pyramids
Spinal cord
2 Action potential in sensory axon
Sensory neuron
1 Sensory endings in skin
Lower motor neuron
7 The upper motor neuron contacts a lower motor neuron in the spinal cord
Neuromuscular junction
8 The lower motor neuron causes contraction of the target skeletal muscle



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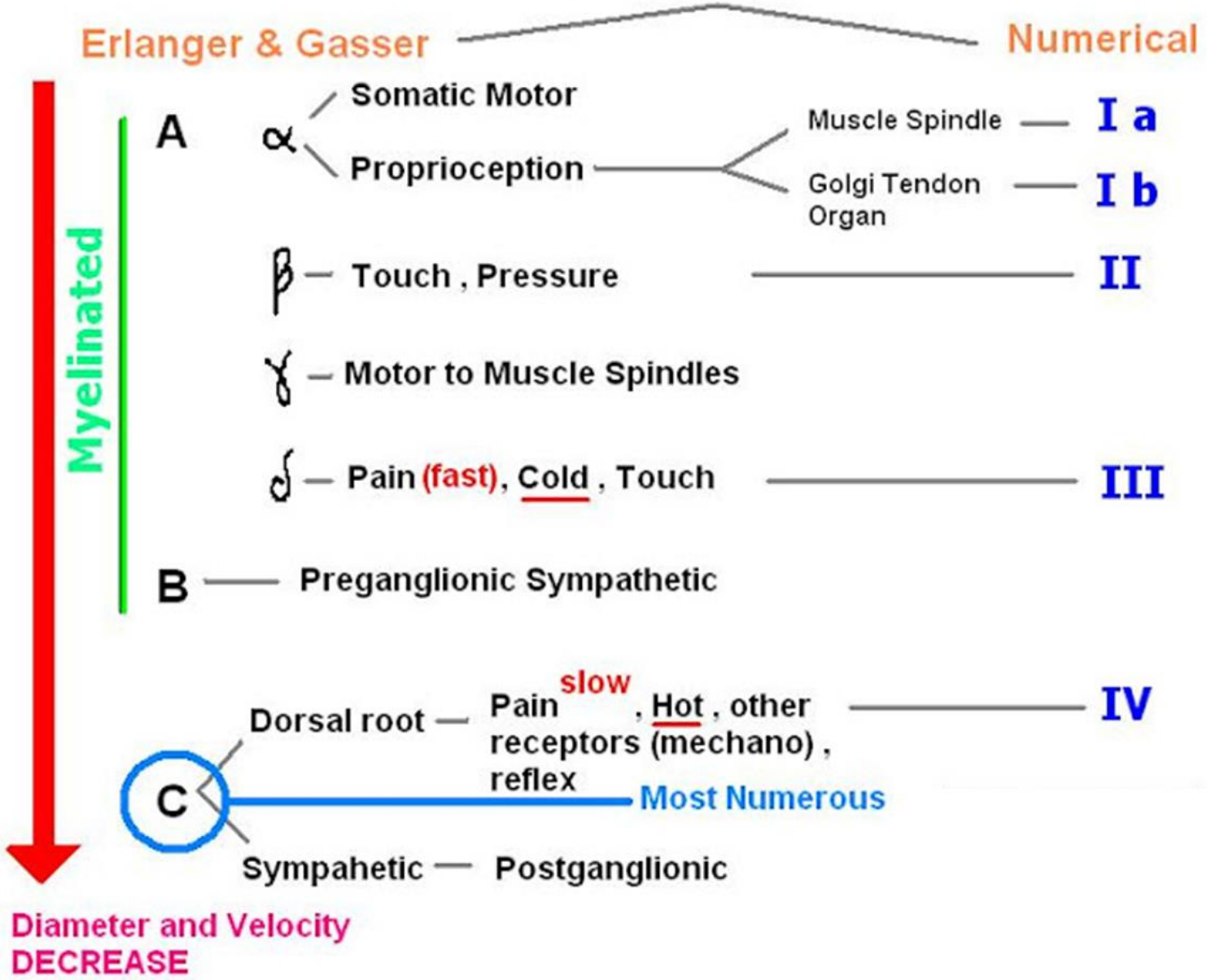
Gate Control Theory of Pain

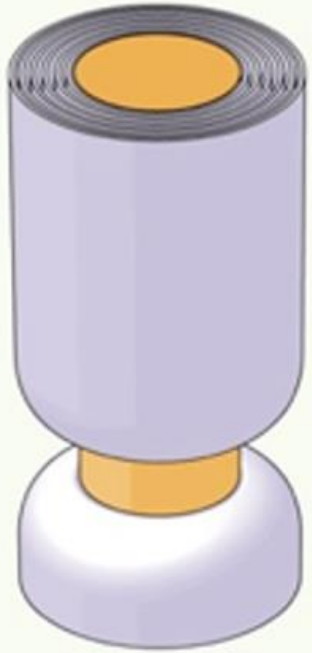


Opioid System

- Endogenous opioids regulate pain.
- An example is feeling relaxed after exercise - caused by the release of these natural chemicals endorphins.

Classification of Nerve fibers



Axons from skin	Aα	Aβ	Aδ	C
Axons from muscles	Group I	II	III	IV
				
Diameter (μm)	13–20	6–12	1–5	0.2–1.5
Speed (m/sec)	80–120	35–75	5–30	0.5–2
	270 mph	**80-170 mph**	**10-70 mph**	**1 mph**
Sensory receptors	Skeletal muscle proprioceptors	Skin mechanoreceptors	Nociceptors, thermoreceptors	
Motor	Fast twitch Skeletal muscle	Slow twitch Skeletal muscle	ANS preganglionic	ANS postganglionic

Electrical Stimulation

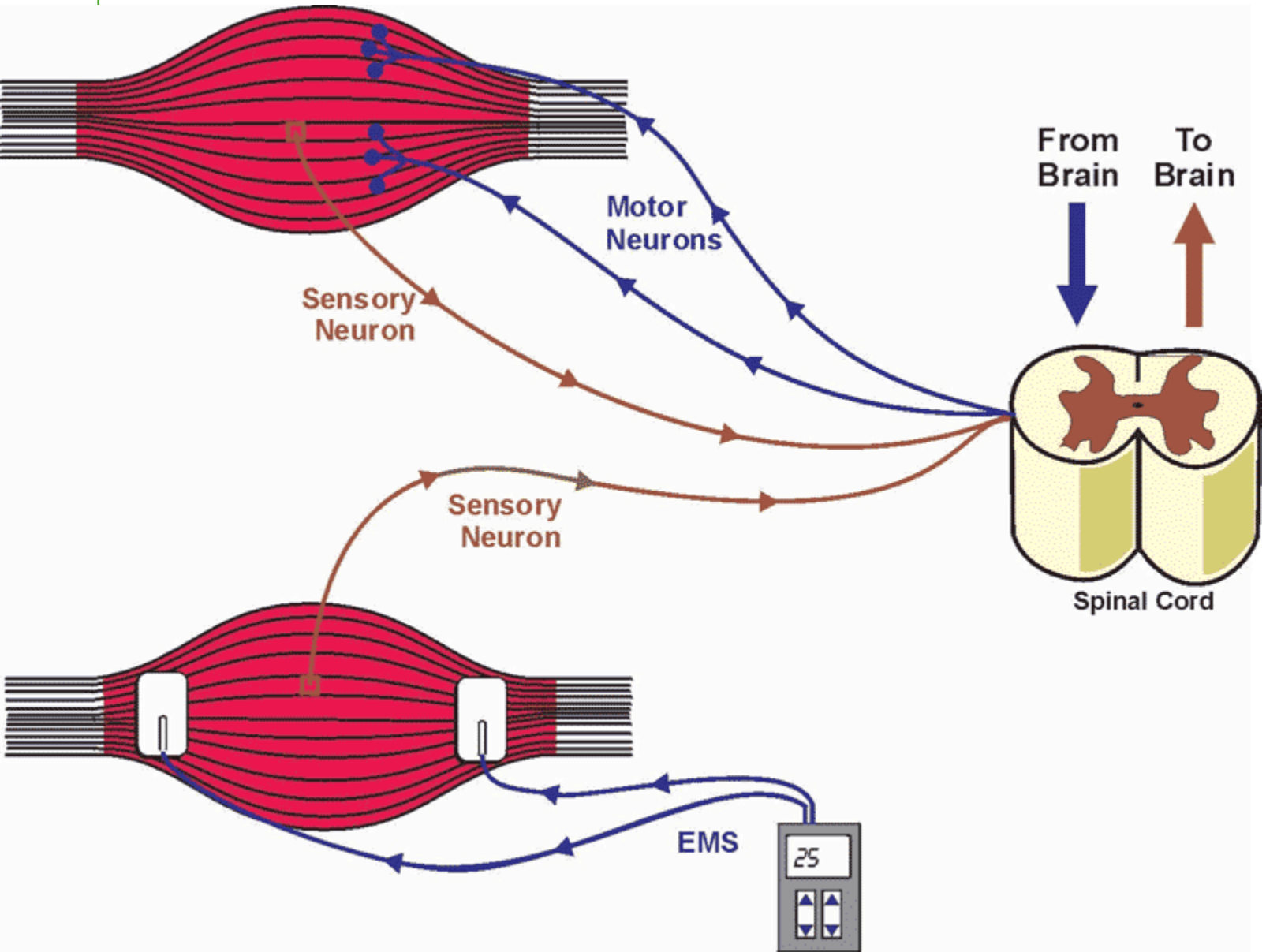
Polarised - A nerve is polarised (ready to discharge) in it’s resting state.

Depolarised - An action potential depolarises the membrane and continues the transmission along the nerve.

Polarised - The nerve then resets to being in a polarised state.

Stimulating the nerve ten times a second, creates 10 action potentials a second.

Electrical stimulation



- The focus of electrical stimulation is the use of electric currents to initiate action potentials in nerves.

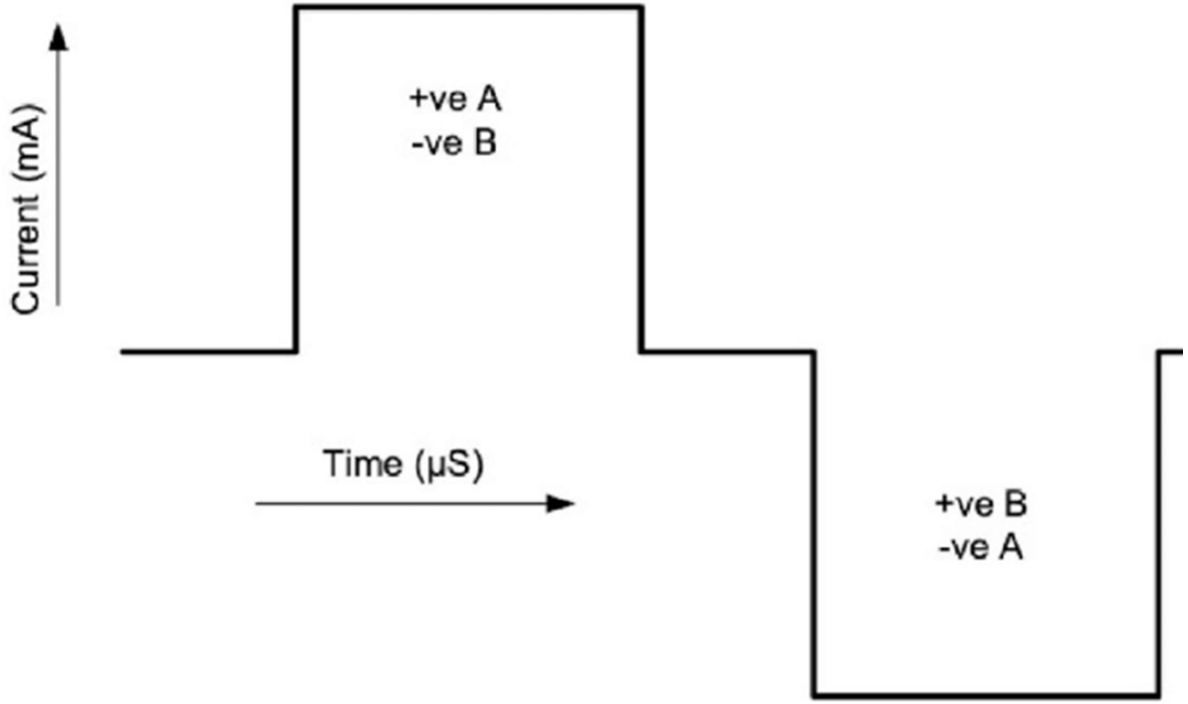
	Muscle Strengthening	Muscle Activation	Pain Relief
TES	**NMES**	**FES**	**TENS**
Therapeutic Electrical Stimulation	Neuromuscular Electrical Stimulation	Functional Electrical Stimulation	Transcutaneous Electrical Nerve Stimulation
Use of electricity to drive a desired nerve response for therapy.	Electricity applied across the surface of the skin over intact peripheral nerve evokes an action potential in the nerve fiber which causes an exchange of ions to drive the muscle to contract.	Application of electrical stimulus to a paralyzed nerve or muscle to restore or achieve function. Also refers to orthotic substitution (Bioness L300).	Pain modulation by exciting peripheral nerves. Common Types: • Sensory • Motor • Noxious

TENS: Pain Relief

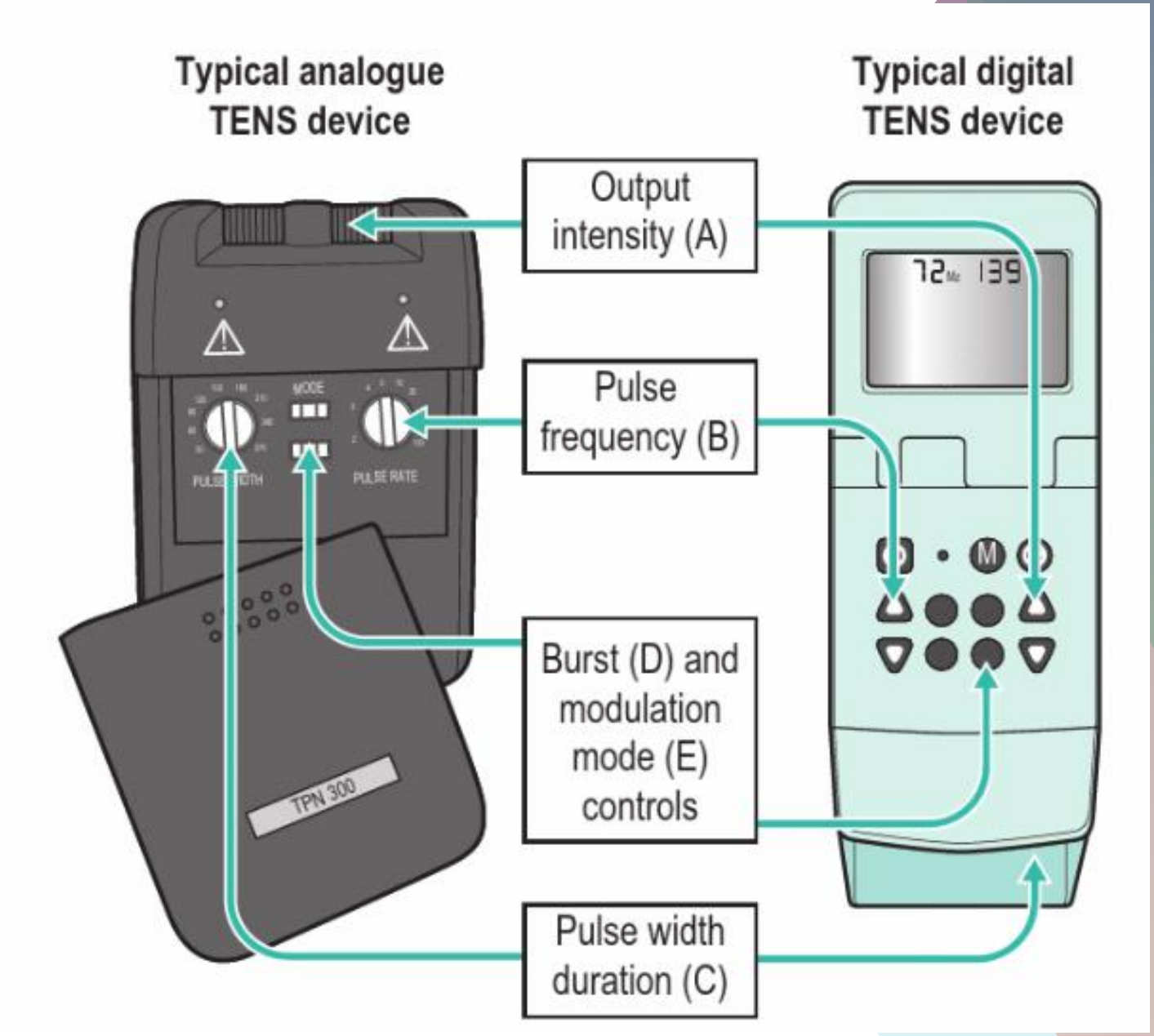
- Electrical stimulation via the skin which works with the pain gate / opioid system to relieve pain.
- Non-invasive & non-pharmacological
- Portable
- Patients can be taught to self-administer

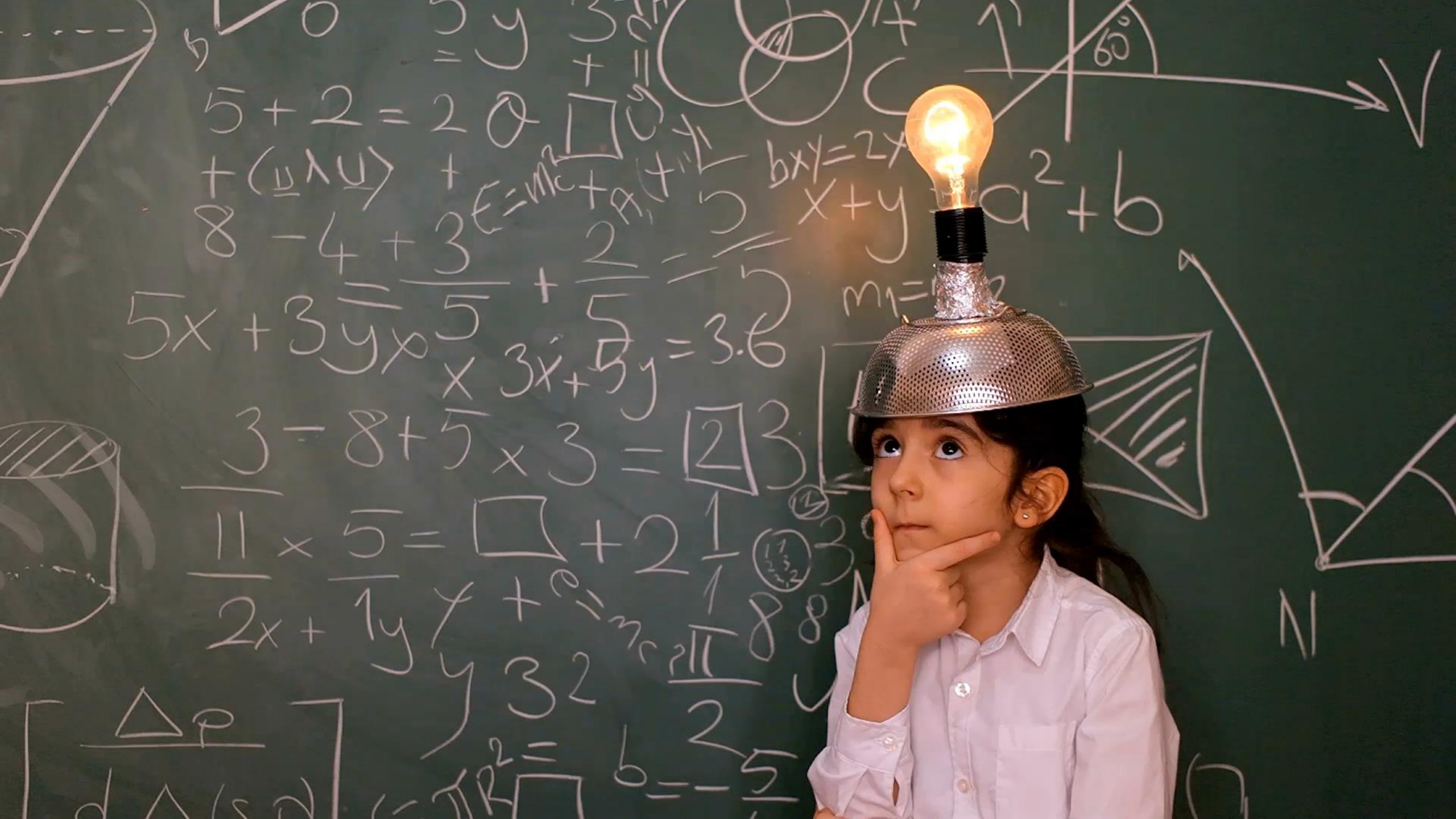
TENS: Pain Relief

- Biphasic, asymmetrical, square wave with zero net DC component:
  - amplitude 0-80mA
  - pulse frequency 1-200 pulses per second
  - pulse width 40 - 50 microseconds.
- Burst mode 2-3 pulses per second
- Modulation mode (E) – less regular to reduce accommodation effects



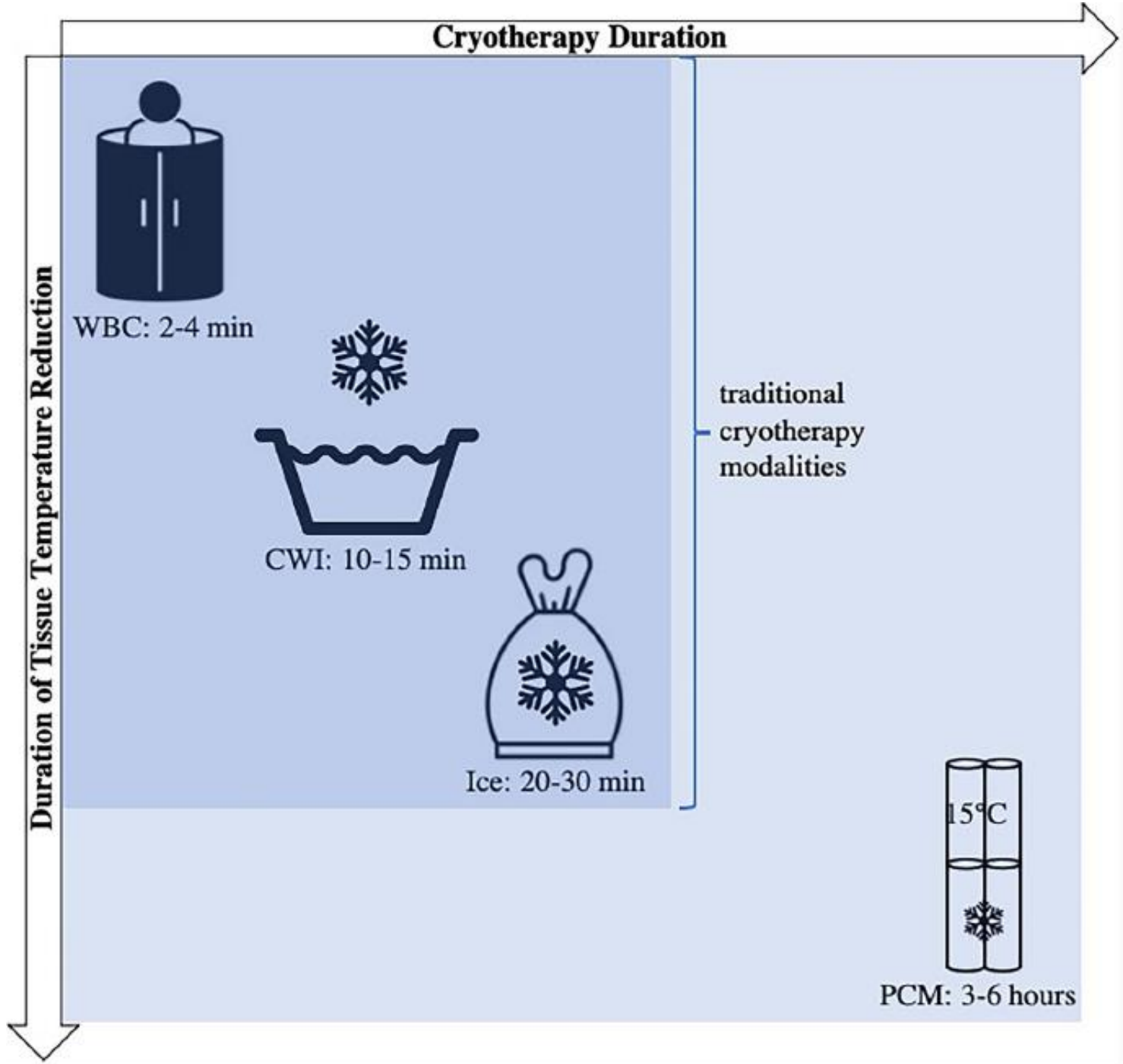
slow rate (around 10 Hz) - slow tickling sensation
high rate (100 Hz +) - continuous buzzing sensation


Typical analogue TENS device
Typical digital TENS device
Output intensity (A)
Pulse frequency (B)
Burst (D) and modulation mode (E) controls
Pulse width duration (C)
MODE
PULSE RATE
TPN 300
72 139


5 + 2 = 20
8 − 4 + 3
5x + 3y
3 − 8 + 5 × 3 = 23
11 × 5 = □ + 2 1/1
2x + 1y
E=mc²
x + y
a² + b
60°

What does cryotherapy do?

- Cryotherapy uses cold and cooling agents for therapeutic management.
- There are various forms, crushed ice, cryocuffs, gel packs, ice submersion, cryosprays, whole body cryotherapy (etc).
- Theoretically cooling the tissues results in vasoconstriction and reduces the metabolic rate of tissues.
- Whilst it is often suggested that there may be faster recovery times from acute injury, delayed onset muscle soreness – evidence is inconclusive.
- Cryotherapy has an analgesic effect by reducing the nerve conduction velocity (Chesterton 2002; Algafly 2007).


Cryotherapy Duration
Duration of Tissue Temperature Reduction
WBC: 2-4 min
CWI: 10-15 min
Ice: 20-30 min
traditional cryotherapy modalities
15°C
PCM: 3-6 hours

Cryotherapy for recovery?

- Cryotherapy has been considered to reduce the metabolic rate / stress by reducing nicotinamide adenine dinucleotide phosphate oxidase (NOX) and reducing mitochondrial reactive oxygen species (ROS) activity).
- There is no conclusive evidence to support the use of ice post-injury in managing inflammation.
- The primary goal of cryotherapy should be to maintain a reduction in intramuscular temperature for as long as possible in the immediate stages following both injury and exercise to hinder the proliferation of secondary damage.

Table 2 Summary of evidence for accelerated recovery following injury and exercise from different types of cryotherapy in humans

	Ice	Whole body cryotherapy	Cold water immersion
Injury	Equivocal	No data	No data
Soreness	✗	✓	✓
Strength	✗	✗	✗
Functional Recovery	No data	✗/✓	✗/✓
Inflammation	No data	✗/✓	✗
Metabolism	Equivocal	No data	Indirect evidence

✗Indicates no direct evidence

✓Indicates evidence to support finding

More research is required on the use of other forms of cryotherapy (such as whole body cryotherapy or cold immersion).

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Cryotherapy Risks

- Cryotherapy risks include frostbite from reducing skin temperature to 20 degrees C, or compression of superficial nerves where an ice pack may be placed.
- The general evidence basis for cryotherapy is mixed.

Cryotherapy Summary

The best use of cryotherapy is to provide pain relief following soft tissue injury, and there may be some tentative evidence for consideration in post exercise soreness.

SDLs

Two systematic reviews have been uploaded on Weblearn.

P

Protect



Protect your injury from further damage, for example, by using a support or splint.

R

Rest



Rest your injury for the first two to three days. You may need to use crutches if you've injured your leg and you want to remain mobile. Then reintroduce movement gradually so you don't delay your recovery by losing muscle strength.

I

Ice



Ice the painful area with a cold compress such as ice or a bag of frozen peas wrapped in a towel. This will help reduce swelling and bruising. Do this for 15 to 20 minutes every two to three hours. Don't apply ice directly to your skin as it can damage it.

C

Compress



Compress the injured area with an elastic bandage or elasticated tubular bandage to help limit swelling and movement. But don't leave the bandage on while you sleep.

E

Elevate



Elevate your injury by resting it above the level of your heart and keep it supported. This could mean lying on the sofa with your foot on some cushions if you've injured your leg.

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O-L

Optimal Loading



While rest may be helpful in the short term, continued rest may lead to joint stiffness, muscle weakness and poor balance. Optimal loading is doing as much as you feel comfortable doing, not pushing through pain.

I

Ice



Ice packs or a bag of frozen peas wrapped in a towel applied to the injured area for 15-20 minutes can reduce swelling and pain. Don't apply ice directly to your skin as it can cause damage.

C

Compression



Compress the injured area by bandaging it to support your injury and help reduce swelling. But don't leave the bandage on while you sleep.

E

Elevate



Elevate the knee (if you can) above the level of your heart to control swelling. Keep the area supported and try to keep it elevated as much as possible until the swelling goes down.

…but there are **Limitations** to using these

Limited benefit of **ice (cryotherapy)**, with potential to harm and reduce healing rates

No accounting for both **acute** and **subacute** stages of healing, which require **different conditions for optimal recovery**

Failure to acknowledge the **influence** of both:

- **Patient education about treatments**
- **Psychological factors of recovery**

Multiple acronyms have been developed over time to remind clinicians how to advise the appropriate aftercare of their patient's injury…




PEACE AND LOVE

PEACE & LOVE

Immediately after injury…
Do no harm and let **PEACE** guide your approach

Protection

Unload and restrict movement for 1-3 days to prevent distension of injured fibres, and reduce the risk of re-injury

Elevation

Elevate higher than the heart where possible to promote interstitial fluid flow out of tissues

Avoid NSAIDs

NSAIDs may negatively affect long-term tissue healing, especially when higher doses are used

Compression

Use bandaging or taping to reduce swelling, and to limit-intra-articular oedema and tissue haemorrhage

Education

Advise your patient to avoid unnecessary treatments, such as acupuncture, manual therapy and electrotherapy

After the first few days…
Get moving and give your soft tissues some **LOVE**

Loading

Add mechanical stress early and resume normal activities as pain allows, promoting repair, remodelling and building tissue tolerance and capacity

Optimism

Give your patient optimistic expectations of their injury, as catastrophisation, depression and fear can represent barriers to adequate recovery

Vascularise

Gradually increase pain-free aerobic activity to boost motivation and increase blood flow to injured structures, improving physical function and decreasing analgesia required

Exercise

Restore strength, stability and proprioception early, but avoid pain and use it as a guide for exercise progression

Jeremy Telford FY2, 10th May 2021

Adapted from Dubois and Esculier (2019):

Dubois, B. and Esculier, J., 2019. Soft-tissue injuries simply need PEACE and LOVE. *British Journal of Sports Medicine*, 54(2), pp.72-73.

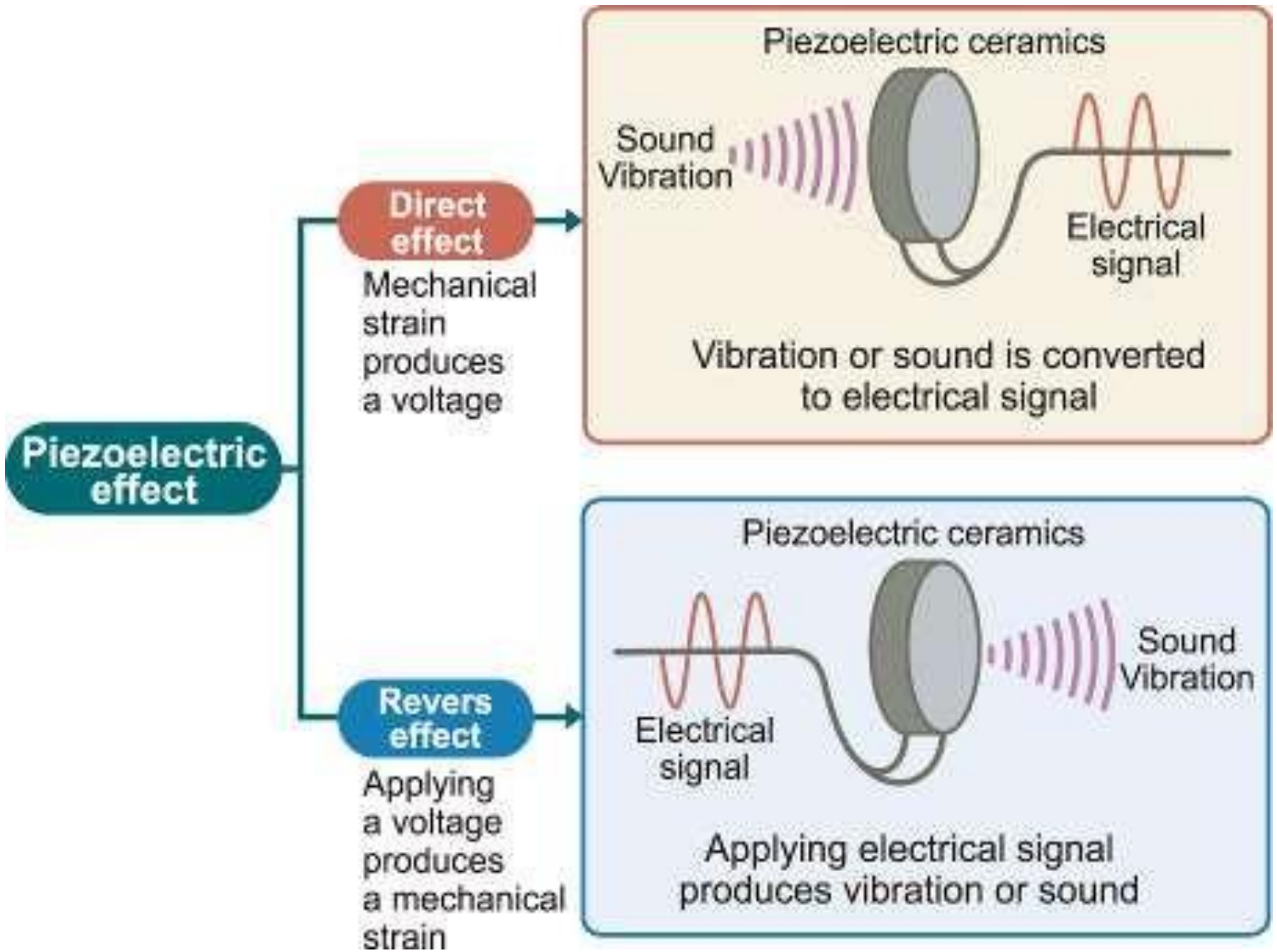
Electrotherapy: Therapeutic Ultrasound



Introduction

- Ultrasound is a type of sound wave that transmits energy by alternately compressing and rarefying material.
- Ultrasound is sound with a frequency greater than 20,000 cycles per second[Hz].
- Humans can hear sound with a frequency of 16 to 20,000Hz, sound greater than this is known as ultrasound.
- Therapeutic ultrasound has a frequency between 0.5 and 5.0 megahertz (MHz) (1-3 MHz) to maximize energy absorption at a depth of 3 to 5 cm of soft tissue.

Production of Ultrasound



The Transducer



Ultrasound Treatment Parameters

- Intensity
  - Pulsed mark : space ratio
- Reflection of ultrasound
- Transmission of ultrasound
- Attenuation of ultrasound
- Ultrasonic fields
- Coupling media
- Duty Cycle



Intensity

- It determines the strength of an US beam
- It is the rate at which energy is delivered per unit
- Watts/square centimeter (W/cm^2)
- The greater intensity the greater temperature rise
- PT – 0.25 to 2.0 W/cm^2 (therapeutic application)

Reflection of ultrasound

Refraction is mainly the deflection from its original path.



An ultrasonic wave travelling through one medium encounters, another medium which will not transmit it , reflection takes place.



There will always be some reflection at each interface that the ultrasound beam encounters.



Acoustic impedance: which is ratio between the reflected and transmitted at an interface.

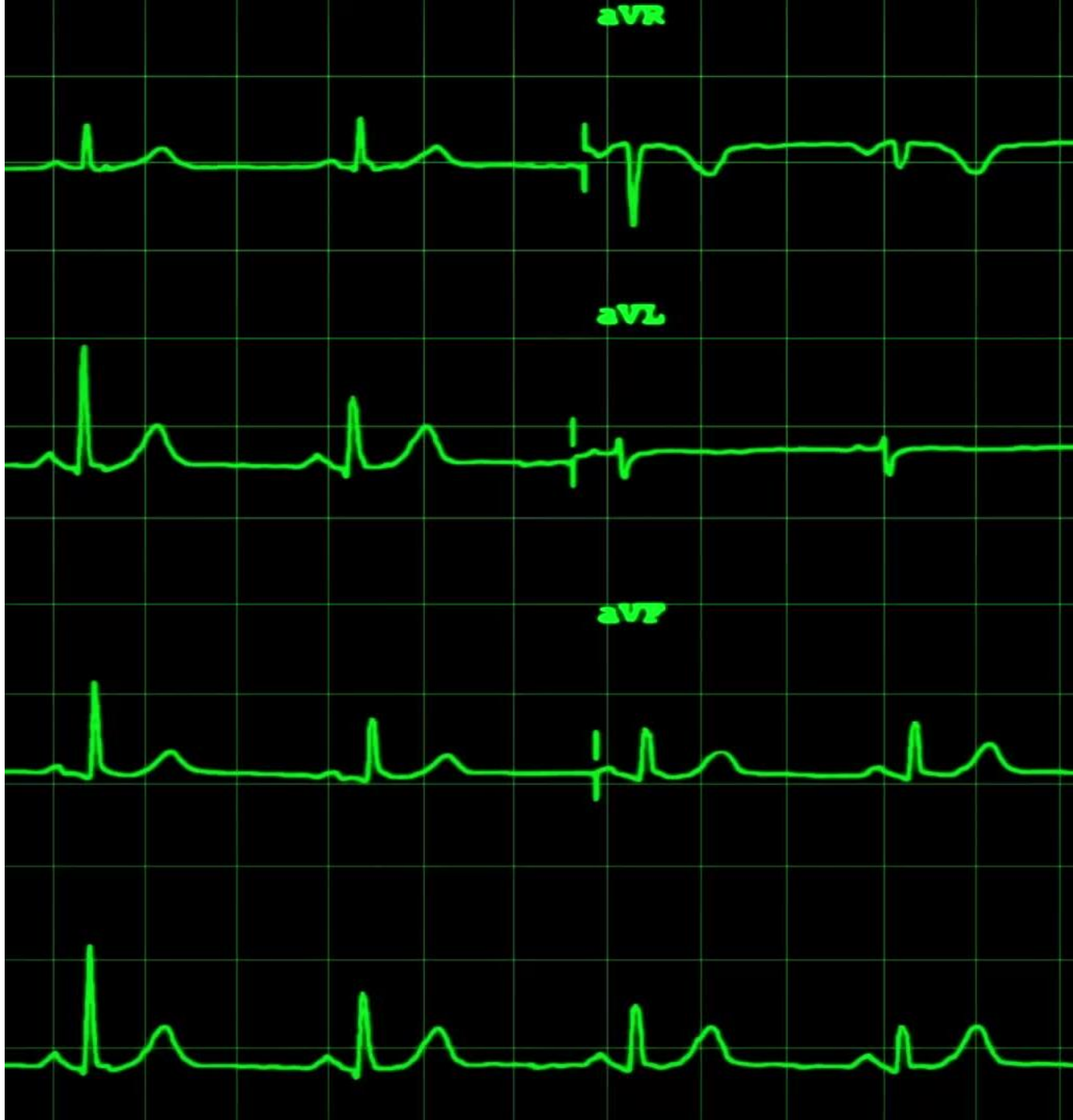
Attenuation of Ultrasound

Attenuation is a gradual reduction in the intensity of ultrasonic beams once it has left the treatment head.

Factors that contribute to attenuation –

- Absorption – ultrasound is absorbed and converted to heat within tissues as a thermal effect.
- Scatter – the cylindrical ultrasonic beam is deflected from its path by reflection at interfaces, bubbles, or particles in its path.

Therefore the US beam intensity decreases with depth.



Major Therapeutic US Frequencies:

- 3 MHz – most of the energy is absorbed within a depth of 1 to 2 cm

- 1 MHz – absorption in deeper tissue

  - deeper than 2 cm from the skin surface (2-5cm)

Coupling Media

Ultrasonic waves are not transmitted by air, thus some couplant which does transmit them must be interposed between the treatment head (transducer) and the patient's skin.

Duty Cycle

For continuous US waves, constant intensity

- Pulsed wave, intensity is intermittently interupted.
- Pulsed waves are further characterized by specifying what fraction of time the wave is present over one pulse period. This fraction is called duty cycle. Duty cycle = duration of pulse /pulse period interrupted

Testing the Apparatus

- Placing the treatment head just below the water surface in a suitable container and observing the disturbance (ripples) which appears.
- The apparatus should be on and off with the treatment head below the water.
- This, and similar methods, only indicate the presence of an output.

Dosage in Acute condition

- In any acute conditions treatment applied cautiously to prevent exacerbation of symptoms.
- Initial stage - Low dose 0.25 to 0.5 Wcm^2 Time 2–3 minutes
- Progression unnecessary if condition improves
- It may also possible to select different pulsed mark to space ratio 1 : 7 for very acute 1 : 1 for less acute.

Dosage in Chronic Conditions

- Chronic condition may be treated with pulsed or continuous mode.
- The maximum intensity of ultrasound which should be used is that which produces mildly perceptible warmth.
- This usually occurs around 2 W/cm2. Initially low dose is tried. Intensity 0.8 W/cm2, time 4 minutes.
- Maximum dose of ultrasound: 2 Watts/cm2 for 8 minutes.

Thermal Effects

- Increase collagen tissue extensibility
- Alterations in blood flow
- Changes in nerve conduction velocity
- Increased pain threshold
- Increase enzymatic activity
- Changes in contractile activity of skeletal muscle

Non – Thermal Effects

- **CAVITATION** – The vibrational effect on gas bubbles by an US beam.

The oscillatory activity of highly compressible bodies within the tissues such as gas or vapour-filled voids.

- **Stable Cavitation** - Stable cavitation occurs when bubbles oscillate to and fro within the ultrasonic pressure waves but remain intact.

- **Microstreaming** is the unidirectional movement of fluids along the boundaries of the cell. Due to microstreaming, the permeability of the cell membrane and the direction of movement of molecules into the cells is influenced.

- **Unstable/Transient Cavitation** - This occurs when the volume of the bubbles changes rapidly and then collapses.

APPLICATION TECHNIQUE: WATER BATH IMMERSION

- Used when contour of the body part does not fit the size of the ultrasound head you are using
- Use a plastic or ceramic container, not metal
- Use room temperature water.
- Hold sound head at a distance of 0.5 to 1 cm from body part and wipe any bubbles off the sound head periodically



Image from http://www.hsiehtherapies.com/CustomImages/Underwater_ultrasound_treatment.jpg

Dangers

- Burns: If a continuous beam is used and is allowed to remain stationary, excess heat can accumulate in the tissues and eventually lead to burns.

- Cavitation: Especially unstable cavitation is dangerous and has been described previously. (Especially with standing/stationary waves)

- Overdose: Excessive dose may cause an exacerbation of symptoms. (Know the maximum dosage and number of sessions of your US equipment).

Evidence for US

More evidence is required to support the use of ultrasound, there are some studies which suggest a potential benefit – but lack strength when compared to placebo effects.

If Ultrasound is to be used, it should be as an adjunct to other therapy, and relevance of tissue depth should also be considered.

Future research may be required with improvements in US technology.

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